	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current	771 - flash_fwd_kernel	(512, 1, 1)x(128, 1, 1)	46.14 ms	64,334,031	0 - NVIDIA GeForce RTX 3090	1.39 Ghz	[126037] python3.12	

Summary
Details
Source
Context
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GPU Speed Of Light Throughput

All

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	40.74	Duration [ms]	46.14
Memory Throughput [%]	17.60	Elapsed Cycles [cycle]	64334031
L1/TEX Cache Throughput [%]	18.69	SM Active Cycles [cycle]	57377907.54
L2 Cache Throughput [%]	17.60	SM Frequency [Ghz]	1.39
DRAM Throughput [%]	0.67	DRAM Frequency [Ghz]	9.49

Latency Issue This workload exhibits low compute throughput and memory bandwidth utilization relative to the peak performance of this device. Achieved compute throughput and/or memory bandwidth below 60.0% of peak typically indicate latency issues. Look at Scheduler Statistics and Warp State Statistics for potential reasons.

Key Performance Indicators

GPU Throughput

Compute (SM) [%]
Memory [%]
0.0
10.0
20.0
30.0
40.0
50.0
60.0
70.0
80.0
90.0
100.0
Speed Of Light (SOL) [%]

Compute Throughput Breakdown

SM: Pipe Tensor Cycles Active [%]	40.74
SM: Issue Active [%]	13.73
SM: Inst Executed [%]	13.73
SM: Inst Executed Pipe Lsu [%]	11.47
SM: Mio2rf Writeback Active [%]	7.96
SM: Pipe Fma Cycles Active [%]	6.45
SM: Inst Executed Pipe Xu [%]	5.73
SM: Pipe Fmaheavy Cycles Active [%]	5.13
SM: Pipe Alu Cycles Active [%]	4.98
SM: Mio Inst Issued [%]	3.93
SM: Mio Pq Write Cycles Active [%]	0.64
SM: Mio Pq Read Cycles Active [%]	0.64
SM: Inst Executed Pipe Adu [%]	0.32
SM: Inst Executed Pipe Cbu Pred On Any [%]	0.00
SM: Inst Executed Pipe Uniform [%]	0.00
SM: Memory Throughput Internal Activity [%]	0
IDC: Request Cycles Active [%]	0
SM: Instruction Throughput Internal Activity [%]	0
SM: Inst Executed Pipe Tex [%]	0
SM: Pipe Fp64 Cycles Active [%]	0
SM: Inst Executed Pipe Ipa [%]	0

Memory Throughput Breakdown

L2: T Sectors [%]	17.60
L2: Lts2xbar Cycles Active [%]	17.59
L1: Data Pipe Lsu Wavefronts [%]	16.68
L1: Lsu Writeback Active [%]	15.92
L1: Lsuin Requests [%]	11.47
L1: M Xbar2l1tex Read Sectors [%]	10.20
L1: Data Bank Reads [%]	7.64
L2: D Sectors [%]	4.47
L2: Xbar2lts Cycles Active [%]	4.41
GPU: Compute Memory Access Throughput Internal Activity [%]	4.41
L2: T Tag Requests [%]	4.40
L1: M L1tex2xbar Req Cycles Active [%]	1.28
L1: Data Bank Writes [%]	1.28
DRAM: Cycles Active [%]	0.67
DRAM: Dram Sectors [%]	0.57
L2: D Sectors Fill Device [%]	0.27
L1: Texin Sm2tex Req Cycles Active [%]	0.04
L2: D Sectors Fill Sysmem [%]	0.00
L1: Tex Writeback Active [%]	0
L2: D Atomic Input Cycles Active [%]	0
L1: F Wavefronts [%]	0
L1: Data Pipe Tex Wavefronts [%]	0
GPU: Compute Memory Request Throughput Internal Activity [%]	0

Floating Point Operations Roofline

100
10
1
0.1
Performance [FLOP/s] (1 = 1e+12)
0.1
1
10
100
1,000
10,000
100,000
Arithmetic Intensity [FLOP/byte]

Floating Point Operations Roofline (Double Precision)

10
1
0.1
Performance [FLOP/s] (1 = 1e+12)
0.01
0.1
1
10
100
Arithmetic Intensity [FLOP/byte]

Floating Point Operations Roofline (Half Precision)

10
1
Performance [FLOP/s] (1 = 1e+13)
1
10
100
1,000
10,000
Arithmetic Intensity [FLOP/byte]

Floating Point Operations Roofline (Single Precision)

100
10
1
0.1
Performance [FLOP/s] (1 = 1e+12)
1
10
100
1,000
10,000
100,000
1e+06
1e+07
Arithmetic Intensity [FLOP/byte]

Tensor Core Operations Roofline

10
1
0.1
Performance [OP/s] (1 = 1e+14)
1
10
100
1,000
10,000
100,000
1e+06
1e+07
1e+08
1e+09
Arithmetic Intensity [OP/byte]

	# Operations	# Operations / Cycle	# Operations / s	Peak %	eak Operations / Cycle	Peak Operations / s
Src:bf16 Dst:fp32 Sparsity:	2199023255552	34210.71	47664396766120.84	81.49	41984	58494600664754.16
Src:bf16 Dst:fp32 Sparsity:	0	0	0	0	83968	116989201329508.31
Src:fp16 Dst:fp16 Sparsity:	0	0	0	0	83968	116989201329508.31
Src:fp16 Dst:fp16 Sparsity:	0	0	0	0	167936	233978402659016.62
Src:fp16 Dst:fp32 Sparsity:	0	0	0	0	41984	58494600664754.16
Src:fp16 Dst:fp32 Sparsity:	0	0	0	0	83968	116989201329508.31
Src:fp64	0	0	0	0	289.54	403411039067.27
Src:int1	0	0	0	0	1343488	1871827221272133
Src:int4 Sparsity:off	0	0	0	0	335872	467956805318033.25
Src:int4 Sparsity:on	0	0	0	0	671744	935913610636066.50
Src:int8 Sparsity:off	0	0	0	0	167936	233978402659016.62
Src:int8 Sparsity:on	0	0	0	0	335872	467956805318033.25
Src:tf32 Dst:fp32 Sparsity:	0	0	0	0	20992	29247300332377.08
Src:tf32 Dst:fp32 Sparsity:	0	0	0	0	41984	58494600664754.16

PM Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over its runtime.

Maximum Sampling Interval [us]	16	# Pass Groups	2
Maximum Buffer Size [Mbyte]	29.03	-	-

0s
ms
+5ms
+10ms
+15ms
+20ms
+25ms
+30ms
+35ms
+40ms
+45ms
Overview
Average Active Warps Per Cy
8 warp
0
Total Active Warps Per Cycle
656 warp
0
Blocks Launched
30 block
0
SM Active Cycles
22.3k cycle
0
Executed Ipc Active
0.65 inst/cycle
0
SM
SM ALU Pipe Throughput
100 %
0
SM FMA Light Pipe Throughp
100 %
0
SM FMA Heavy Pipe Through
100 %
0
SM Tensor Pipe Throughput
100 %
0
SM XU Pipe Throughput
100 %
0
DRAM
DRAM Throughput
100 %
0
DRAM Read Bandwidth
100 %
0
DRAM Write Bandwidth
100 %
0
L2 Cache
L2 Throughput
100 %
0
L2 Hit Rate
100 %
0
L1 Cache
L1 Throughput
100 %
0
L1 Writeback Throughput
4.09k cycle
0
L1 Hit Rate
100 %
0
L1 Wavefronts (Data)
4.44k
0
Workload Execution
flash_fwd_kernel

Compute Workload Analysis

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	0.55	SM Busy [%]	40.74
Executed Ipc Active [inst/cycle]	0.62	Issue Slots Busy [%]	13.73
Issued Ipc Active [inst/cycle]	0.62		

Pipe Utilization (% of elapsed cycles)

0.0
25.0
50.0
75.0
100.0
Tensor (All)
Tensor (FP)
FMA
ALU
FP64
Tensor (INT)
Utilization [%]

Pipe Utilization (% of peak instructions executed over elapsed cycles)

0.0
25.0
50.0
75.0
100.0
Tensor (FP)
LSU
FMA
XU
ALU
ADU
CBU
Uniform
FMA (FP16)
FP64 (DMMA)
FP64 (FP64)
TEX
Tensor (INT)
Utilization [%]

Pipe Utilization (% of active cycles)

0.0
25.0
50.0
75.0
100.0
Tensor (All)
Tensor (FP)
FMA
ALU
FP64
Tensor (INT)
Utilization [%]

Pipe Utilization (% of peak instructions executed over active cycles)

0.0
25.0
50.0
75.0
100.0
Tensor (FP)
LSU
FMA
XU
ALU
ADU
CBU
Uniform
FMA (FP16)
FP64 (DMMA)
FP64 (FP64)
TEX
Tensor (INT)
Utilization [%]

Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each memory unit.

Memory Throughput [Gbyte/s]	6.09	Mem Busy [%]	17.60
L1/TEX Hit Rate [%]	0.00	Max Bandwidth [%]	17.59
L2 Hit Rate [%]	98.47	Mem Pipes Busy [%]	11.47
L2 Compression Input Sectors [sector]	0	Local Memory Spilling Requests	4096
L2 Compression Ratio	0	Local Memory Spilling Request Overhead [%]	100
L2 Compression Success Rate [%]	0	L2 Persisting Size [Mbyte]	1.18

Local Memory Usage This workload accesses local memory. Local memory refers to a memory space that is private to each thread. Although it is logically distinct, local memory resides in global memory. This can lead to higher memory latency compared to the usage of registers or shared memory, a decrease in available bandwidth to global memory, and an increase in executed instructions.
The local memory accesses of this workload are largely cached in L1TEX (86.38% of loads, and 81.40% of stores). However, even spilling to L1TEX might lead to high performance penalties compared with staying in registers.
One common reason for the use of local memory is arrays being too large to fit into registers or not being indexed using compile time constants (see the "Local Memory" section of the CUDA Programming Guide. For more advice on how to reduce the usage of local memory refer to the documentation.

Key Performance Indicators

Memory Chart

Values: Throughput Inactivity: Greyed Out

Kernel
40.96 K Inst
Global
0.00 Req
40.96 K Req
4.10 K Inst
Local
2.05 K Req
2.05 K Req
0.00 Inst
Texture
0.00 Req
0.00 Inst
Surface
0.00 Req
0.00 Req
33.59 M Inst
Load Global Store Shared
201.49 M Inst
Shared
201.36 M Req
131.07 K Req
L1/TEX Cache Hit Rate: 0.00 %
372.74 GB/s
375.01 MB/s
372.74 GB/s
0.00 B/s
hared Memory
L2 Cache Hit Rate: 98.47 %
0.00 B/s
0.00 B/s
L2 Compression Ratio: 0.00
258.05 KB
System Memory
0.00 B/s
0.00 B/s
Device Memory
5.68 GB/s
408.86 MB/s
Peer Memory
0.00 B/s
0.00 B/s
333.23 M Req
ICC Hit Rate: 100.00 %
10.70 K Req
235.01 M Req
IMC Hit Rate: 100.00 %
2.63 K Req
0.00 Req
IDC Hit Rate: 0.00 %
GCC Hit Rate: 92.49 %
% Peak
100%
80%
60%
40%
20%
0%

Shared Memory

	Instructions	Requests	Wavefronts	% Peak	Bank Conflicts
Shared Load	32768	32768	805448634	15.28	10906
Shared Load Matrix	201326592	201326592			
Shared Store	131072	131072	131072	0.00	0
Shared Store From Global Lc	33587200	33587200	134348800	0.64	0
Shared Atomic	0	0	0	0	0
Other	-	-	60838070	1.16	1459
Total	235077632	235077632	1000766576	17.08	12365

L1/TEX Cache

	% Peak	Sectors	Sectors/Req	Hit Rate	Bytes	Sector Misses to L2	% Peak to L2	Returns to SM	% Peak to SM
Local Load	0.00	8192	4	86.38	262144	1112	7.65	5490	0.00
Global Load		0	0	0	0				
Global Load To Shared Store (acce	0.64	0	0		0			-	-
Global Load To Shared Store (bypa		537395200	16	-	17196646400	537395200	2.55	-	-
Surface Load	0	0	0	0	0	0	0	0	0
Texture Load	0	0	0	0	0	0	0		
Global Store	0.00	532480	13	1.06	17039360	540672	0.01	-	-
Local Store	0.00	8192	4	81.40	262144			-	-
Surface Store	0	0	0	0	0	0	0	-	-
Global Reduction	0	0	0	0	0	0	0	-	-
Surface Reduction	0	0	0	0	0	0	0	-	-
Global Atomic ALU	0	0	0	0	0	0	0	see above	see above
Global Atomic CAS									
Surface Atomic ALU	0	0	0	0	0	0	0	see above	see above
Surface Atomic CAS									
Loads	0.64	537403392	16.00	0.00	17196908544	537396312	10.20	5490	0.00
Stores	0.00	540672	12.57	2.27	17301504	540672	0.01	-	-
Atomics & Reductions	0	0	0	0	0	0	0	-	-

L2 Cache

	ors	Sectors/Req	% Peak	Hit Rate	Bytes	Throughput	Sector Misses to Devic	ector Misses to Systen	Sector Misses to Peer
L1/TEX Load	37396312	4.00	17.58	98.48	17196681984	372742521515.73	8170668	0	0
L1/TEX Store	540672	3.97	0.02	100	17301504	375014565.77	0	0	0
L1/TEX Atomic ALU	0	0	0	0	0	0	0	0	0
L1/TEX Atomic CAS	0	0	0	0	0	0	0	0	0
L1/TEX Reduction	0	0	0	0	0	0	0	0	0
L1/TEX Total	37936956	4.00	17.60	98.48	17213982592	373117516660.47	8181564	0	0
GCC Total	8064	4	0.00	86.01	258048	5593257.02	1128	0	0
ECC Total	0	-	0	-	0	0	0	-	-

L2 Cache Eviction Policies

	First	Hit Rate	Last	Hit Rate	Normal	Hit Rate	Normal Demote	Hit Rate
L1/TEX Load	0	0	0	0	537396440	98.48	0	0
L1/TEX Store	0	0	0	0	540672	100	0	0
L1/TEX Atomic	0	0	0	0	0	0	-	-
L1/TEX Total	0	0	0	0	537923868	98.48	0	0
GPU Total	37420	53.38	0	0	537944183	98.48	0	0

Device Memory

	Sectors	% Peak	Bytes	Throughput
Load	8188416	0.62	262029312	5679552983.35
Store	589468	0.04	18862976	408859874.48
Total	8777884	0.67	280892288	6088412857.83

Scheduler Statistics

Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Issued Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates poor latency hiding.

Active Warps Per Scheduler [warp]	1.88	No Eligible [%]	84.62
Eligible Warps Per Scheduler [warp]	0.21	One or More Eligible [%]	15.38
Issued Warp Per Scheduler	0.15		

Warps Per Scheduler

GPU Maximum Warps Per Scheduler
Theoretical Warps Per Scheduler
Active Warps Per Scheduler
Eligible Warps Per Scheduler
Issued Warp Per Scheduler
0.0
4.0
8.0
12.0
16.0

Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

Warp Cycles Per Issued Instruction [cycle]	12.24	Avg. Active Threads Per Warp	32.00
Warp Cycles Per Executed Instruction [cycle]	12.24	Avg. Not Predicated Off Threads Per Warp	28.38

Warp State (All Cycles)

0.0
2.0
4.0
6.0
8.0
Warp States
Stall Math Pipe Throttle
Stall Barrier
Stall Wait
Selected
Stall Not Selected
Stall MIO Throttle
Stall Dispatch Stall
Stall No Instruction
Stall Short Scoreboard
Stall Long Scoreboard
Stall Branch Resolving
Stall LG Throttle
Stall IMC Miss
Stall Drain
Stall Misc
Stall Membar
Stall Sleeping
Stall Tex Throttle
Cycles per Instruction

Instruction Statistics

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	2895349760	Avg. Executed Instructions Per Scheduler [inst]	8827285.85
Issued Instructions [inst]	2895370365	Avg. Issued Instructions Per Scheduler [inst]	8827348.67

Executed Instruction Categories

0.0
2,000,000,000.0
4,000,000,000.0
Opcode Categories
Floating Point
Load/Store
Integer
Conversion
Movement
Miscellaneous
Control
Uniform Datapath
Predicate
Executed Warp-Level Instructions/Opcode

Executed Instruction Mix

0.0
500,000,000.0
1,000,000,000.0
1,500,000,000.0
Opcodes
FMUL
HMMA
FADD
LDSM
FSETP
FMNMX
MUFU
F2FP
IMAD
LEA
LDGSTS
SHFL
IADD3
SHF
LDS
MOV
FFMA
FSEL
BRA
BAR
LDGDEPBAR
DEPBAR
ISETP
STS
LOP3
STG
BSYNC
BSSY
SEL
S2R
ULDC
USHF
LDG
IABS
EXIT
UMOV
UIADD3
STL
R2P
LDL
I2F
F2I
CS2R
Executed Warp-Level Instructions/Opcode

NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Topology

The system does not have any NVLink connections.

NVLink Tables

Detailed tables with properties for each NVLink.

Logical NVLink Properties

The system does not have any NVLink connections.

NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

NUMA ID Table

GPU ID	GPU Name	NUMA ID by CPU Affini	CPU Affinity	NUMA ID by Memory Affinity
0	NVIDIA GeForce RTX 3090	0	0-29	0

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	512	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	255	Static Shared Memory Per Block [byte/block]	0
Block Size	128	Dynamic Shared Memory Per Block [Kbyte/block]	49.15
Threads [thread]	65536	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	3.12	Shared Memory Configuration Size [Kbyte]	102.40
Uses Green Context	0	Stack Size	1024
# SMs [SM]	82	# TPCs	41
Enabled TPC IDs	all	-	-

Tail Effect
Est. Speedup: 25.00% A wave of thread blocks is defined as the maximum number of blocks that can be executed in parallel on the target GPU. The number of blocks in a wave depends on the number of multiprocessors and the theoretical occupancy of the kernel. This kernel launch results in 3 full waves and a partial wave of 21 thread blocks. Under the assumption of a uniform execution duration of all thread blocks, this partial wave may account for up to 25.0% of the total runtime of this kernel. Try launching a grid with no partial wave. The overall impact of this tail effect also lessens with the number of full waves executed for a grid. See the Hardware Model description for more details on launch configurations.

Key Performance Indicators

Occupancy

% Occupancy Graphs

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	16.67	Block Limit Registers [block]	2
Theoretical Active Warps per SM [warp]	8	Block Limit Shared Mem [block]	2
Achieved Occupancy [%]	15.69	Block Limit Warps [block]	12
Achieved Active Warps Per SM [warp]	7.53	Block Limit SM [block]	16

Theoretical Occupancy
Est. Local Speedup: 83.33% The 2.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (16.7%) is limited by the number of required registers, and the required amount of shared memory.

Key Performance Indicators

Impact of Varying Register Count Per Thread

125
100
75
50
25
0
% Occupancy
0
8
16
24
32
40
48
56
64
72
80
88
96
104
112
120
128
136
144
152
160
168
176
184
192
200
208
216
224
232
240
248
256
Registers Per Thread

Impact of Varying Block Size

125
100
75
50
25
0
% Occupancy
0
32
64
96
128
160
192
224
256
288
320
352
384
416
448
480
512
544
576
608
640
672
704
736
768
800
832
864
896
928
960
992
1,024
Block Size

Impact of Varying Shared Memory Usage Per Block

125
100
75
50
25
0
% Occupancy
0
3,184
6,368
9,552
12,736
15,920
19,104
22,288
25,472
28,656
31,840
35,024
38,208
41,392
44,576
47,760
50,944
54,128
57,312
60,496
63,680
66,864
70,048
73,232
76,416
79,600
82,784
85,968
89,152
92,336
95,520
98,704
101,888
Shared Memory Per Block

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	57377907.54	Average L1 Active Cycles [cycle]	57377907.54
Average L2 Active Cycles [cycle]	26322842.54	Average SMSP Active Cycles [cycle]	57379543.77
Average DRAM Active Cycles [cycle]	2925961.33	Total SM Elapsed Cycles [cycle]	5270860724
Total L1 Elapsed Cycles [cycle]	5270860724	Total L2 Elapsed Cycles [cycle]	3056013024
Total SMSP Elapsed Cycles [cycle]	21083442896	Total DRAM Elapsed Cycles [cycle]	5255294976

SMs Workload Imbalance
Est. Speedup: 9.59% One or more SMs have a much higher number of active cycles than the average number of active cycles. Maximum instance value is 10.75% above the average, while the minimum instance value is 3.96% below the average.

Key Performance Indicators

SMSPs Workload Imbalance
Est. Speedup: 9.62% One or more SMSPs have a much higher number of active cycles than the average number of active cycles. Maximum instance value is 10.77% above the average, while the minimum instance value is 3.95% below the average.

Key Performance Indicators

L1 Slices Workload Imbalance
Est. Speedup: 9.59% One or more L1 Slices have a much higher number of active cycles than the average number of active cycles. Maximum instance value is 10.75% above the average, while the minimum instance value is 3.96% below the average.

Key Performance Indicators

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	57377907.54	55105856	64285446	4704988418
SMSP Active Cycle	57379543.77	55114333	64307898	18820490356
L1 Active Cycles	57377907.54	55105856	64285446	4704988418
L2 Active Cycles	26322842.54	26111992	26631476	1263496442
DRAM Active Cycle	2925961.33	2923424	2927888	35111536

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	12623872	Branch Efficiency [%]	99.98
Branch Instructions Ratio [%]	0.00	Avg. Divergent Branches [branches]	6.24

Warp Stall Sampling (All Samples)

Location	Value	Value (%)
0x7df059281870 in flash_fw...	220,762	10
0x7df059281760 in flash_fw...	135,166	6
0x7df059281e20 in flash_fw...	103,364	4
0x7df0592833c0 in flash_fw...	46,204	2
0x7df059283710 in flash_fw...	30,220	1

Most Instructions Executed

Location	Value	Value (%)
0x7df059283fc0 in flash_fw...	4,192,256	0
0x7df059283fb0 in flash_fw...	4,192,256	0
0x7df059283fa0 in flash_fw...	4,192,256	0
0x7df059283f90 in flash_fw...	4,192,256	0
0x7df059283f80 in flash_fw...	4,192,256	0

To customize your report even further, you might want to learn about custom sections and writing your own rules.
You might also want to consider adding individual metrics.